

MTH166:DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS

L:3 T:1 P:0 Credits:4

Course Outcomes: Through this course students should be able to

CO1 :: define and distinguish between different types of differential equations.

CO2 :: understand the use of different methods for solving differential equations.

CO3 :: interpret the solutions of second-order partial differential equations, including the heat, wave, and Laplace equations.

CO4 :: apply the concepts associated with derivatives of vector fields, such as gradient, divergence, and curl, to solve various problems.

CO5 :: compute line, surface, and volume integrals using various theorems of vector calculus.

Unit I

Ordinary differential equations : exact equations, equations reducible to exact equations, equations of the first order and higher degree, Clairaut's equation

Unit II

Differential equations of higher order : introduction to linear differential equation, Solution of linear differential equation, linear dependence and linear independence of solution, method of solution of linear differential equation- Differential operator, solution of second order homogeneous linear differential equation with constant coefficient, solution of higher order homogeneous linear differential equations with constant coefficient

Unit III

Linear differential equation : solution of non-homogeneous linear differential equations with constant coefficients using operator method, method of variation of parameters, method of undetermined coefficient, solution of Euler-Cauchy equation, simultaneous differential equations by operator method

Unit IV

Partial differential equation : introduction to partial differential equation, method of Separation of Variables, solution of wave equation, solution of heat equation, solution of Laplace equation

Unit V

Vector calculus I : limit, continuity and differentiability of vector functions, length of space curve, motion of a body or particle on a curve, gradient of a scalar field and directional derivatives, divergence and curl of vector field

Unit VI

Vector calculus II : line integral, Greens' theorem, surface area and Surface integral, Stokes' theorem, Gauss's divergence theorem

Text Books:

1. ADVANCED ENGINEERING MATHEMATICS by R.K.JAIN, S.R.K. IYENGER, NAROSA PUBLISHING HOUSE

References:

1. HIGHER ENGINEERING MATHEMATICS by DR. B.S. GREWAL, KHANNA PUBLISHERS